Bollinger Band Strategy

Bollinger Band Bot (Factory Bot v1.3, label BOLL) · the first strategy found by our Strategy Factory · updated 15 October 2026

+0.20R
average per trade, 2010–2026 (2021–26: +0.31R)

59%
of trades win (328 trades)

~24
trades a year, across 9 indices

0.05
correlation with Daily Systems (NAS bot: about 0)

Why it should work

When a stock index has an unusually sharp fall, forced and emotional selling tends to overshoot, and the price usually recovers part of the drop within about a week. Large funds can't easily trade this across 9 smaller index CFDs at once — at our size we can.

The rules

- **Markets:** 9 stock indices — US500, US100, US30, GER40, UK100, FRA40, EU50, AUS200, HK50.
- **Buy** when an index closes **below its lower Bollinger band** (the 20-day average minus 2.5 times the 20-day standard deviation). The order is sent 60 minutes after the next daily open, once that market is trading.
- **Sell 5 trading days later.** No take profit.
- **Flat over the weekend:** every position is closed on Friday at 15:30 New York time (20:30 UK; a market that shuts earlier, such as HK50, closes in its last 45 minutes) and reopened at the Monday order time with the same size and stop. If the 5-day exit falls on Friday's close, it simply stays closed. This avoids Monday gaps and FTMO's triple Friday swap.
- **No normal stop.** Testing showed a normal stop sells just before the bounce and wipes out the edge.
- **Disaster stop** 10 times the average daily range below the entry, purely for a true crash. It has almost never been hit.
- One position per index; several indices can be held at once, because sharp falls often hit several together.
- **Risk budget (v1.1):** all open positions together never add up to more than 6 full-size positions. If 6 are already open, new ones get a smaller size. So even if every disaster stop hit on the same day, the most it could lose is 30R ($3,000 at half size) — the worst real day in 16 years was 5R.

Size

- **Risk Per R = $200**, where R = twice the index's average daily range. The bot works out the position size itself.
- A normal losing trade costs about $200–400. The worst single trade in 16 years lost about $1,000 (5R); the worst single day was about −$1,100 (5.6R).
- **First 30 live trades at half size ($100 per R)**, as our Quant Process says.
- Worst case at once: about $35k of positions, roughly $2.3k of margin at FTMO's 1:15 leverage.

How it was tested

Stage	Result
Strategy Factory: 8,832 strategies tested on 28 markets	34 survived the build and confirm periods; 29 passed the final check
Build period (to 2020)	+0.13R per trade
Confirm period 2021–23 (unseen)	+0.32R per trade
Final check 2024–26 (opened once)	+0.30R per trade, likely range above zero
Edge over simply being in the market	About +0.20R per trade better than buying every day with the same exit
Worst drawdown (2010–26)	11.5R (about $2,300 at $200 per R)
Account simulation with the NAS bot ($550) and Daily Systems	Fail odds 16.8% → 16.2%; passes about 6 days sooner

All tests include spread, commission and overnight fees; signals use the daily close and fill at the next open.

Your cTrader backtests (M1 bars, 1 Jan 2021 – 14 Oct 2026, $200 per R, commission 30)

	v1.1 (risk budget)	v1.3, weekend flat = Yes (live)
Net profit	+$5,954	**+$7,291**
2021–23 / 2024–26	+$1,926 / +$4,027	+$2,010 / +$5,282
Max equity drawdown	4.09%	**3.79%**
Largest loss	−$472	−$400
Swap paid	−$273	−$153
Indices positive	7 of 9	7 of 9

The weekend rule was tested against a pass mark written down before the run (keep 95% of the profit, no higher drawdown). It made more and lowered the drawdown, in both periods. Live, at $100 per R, these dollar figures halve.

Live settings

- Attach to a **US500.cash, Daily** chart; the bot trades all 9 indices itself.
- **Risk Per R = 100** (half size); **Flat Over Weekend = Yes**; **Label = BOLL**; everything else at its default.
- **Crash-safe:** the bot saves its to-do list every minute and reloads it after a restart. After restarting cTrader, look for the **R|** line in the log. Avoid restarting between about 23:00 and 00:30 UK, when orders are sent.
- Log lines: **O|** opened, **C|** closed by the 5-day rule, **W|** weekend close or reopen, **X|** skipped.

Switch-off rule (monthly review)

Pause the bot if its last 50 trades average below −0.05R, or if its drawdown passes $3,500 (1.5 times the worst the backtest showed). A paused bot is re-tested before it restarts.

After 30 live trades that pass the monthly review, Risk Per R can go up to $200.
Files: bots/07_Factory/Factory_Bot.cs (the bot), STRATEGY.md, research/factory (the Strategy Factory and test scripts).